

NoteTube Technical Documentation

Project Title: NoteTube - AI-Powered Video Synthesis and Learning Platform
Credit Hours: 4(0-4) **Technologies:** Next.js 15, Supabase, Gemini AI, Tailwind CSS

Chapter No 1: Introduction to the Problem

1.1 Introduction

In the digital age, YouTube has become a primary source of educational content. However, passive video consumption often leads to poor retention. Students and professionals spend hours manually pausing videos to take notes and lack a structured way to test their understanding. NoteTube solves this by automating the transformation of lengthy videos into structured study materials.

1.2 Background

Traditional note-taking is time-consuming. With the advent of Large Language Models (LLMs) and advanced transcription services, there is an opportunity to bridge the gap between video consumption and active learning. NoteTube leverages Google's Gemini AI to synthesize transcripts into logical, hierarchical notes and interactive self-assessment quizzes.

1.3 Purpose

The purpose of this project is to provide a central learning hub where users can:

- Instantly extract content from any educational video.
- Generate AI-summarized notes that highlight key concepts.
- Self-assess through dynamically generated quizzes.
- Track long-term learning progress via analytics.

1.4 Scope

NoteTube is a web-based educational technology (EdTech) tool. Its scope includes transcript extraction via YouTube metadata APIs, AI processing for summarization, a secure user dashboard for history tracking, and a visual analytics engine to monitor study streaks and time allocation.

1.5 Objective

- **Automate Note-Taking:** Eliminate the need for manual transcription.
- **Enhance Retention:** Use AI to create quizzes that force active recall.
- **Personalize Learning:** Provide a history and bookmarking system for targeted review.
- **Gamify Education:** Use streak tracking and progress bars to motivate consistent study habits.

Chapter No 2: Software Requirement Specification (SRS)

2.1 Overall Description

2.1.1 Product Perspectives NoteTube is a standalone SaaS-style web application integrated with Google Gemini (AI) and Supabase (Backend-as-a-Service). It acts as an intermediary between YouTube’s massive content library and the user’s personal study environment.

2.1.2 Product Features

- **Transcript Extraction:** High-speed retrieval of video captions.
- **AI Synthesis:** Context-aware summarization into Markdown format.
- **Interactive Quizzes:** Multiple-choice questions based on video content.
- **Learning Dashboard:** Centralized view of recent activity and saved materials.
- **Real-time Analytics:** Visual representation of learning time and effectiveness.

2.1.3 Design and Implementation Constraints

- **API Limits:** Dependent on Gemini AI and YouTube transcript availability.
- **Next.js 15+ Standards:** Adherence to “App Router” and Server Component architectures.
- **Mobile Responsiveness:** Must remain fully functional on mobile devices using Tailwind CSS.

2.1.4 Assumption and Dependencies

- Users have a stable internet connection.
- Videos processed must have English captions/transcripts available.
- Browser supports modern JavaScript features (Web Vitals).

2.2 System Features

2.2.1 System Feature 1: Video Processing Engine Retrieves YouTube IDs and interacts with the internal `/api/extract-transcript` endpoint to fetch raw text data. #### **2.2.2 System Feature 2: AI Knowledge Synthesizer** Uses a customized prompt engineering layer to transform raw text into formatted study notes via the Gemini Pro 1.5 model.

2.3 External Interface Requirements

2.3.1 User Interfaces Clean, sleek UI using Shadcn/UI components, Framer Motion for micro-animations, and a dark-mode optimized color palette. ####

2.3.2 Hardware Interfaces Standard web display via desktops, tablets, or smart-phones. No specialized hardware required. ##### 2.3.3 Software Interfaces - **OS:** Windows/Linux/macOS/iOS/Android. - **Supabase API:** For authentication and DB operations. - **Gemini SDK:** For AI generation.

2.4 Other Nonfunctional Requirements

2.4.1 Performance Requirements

- AI responses should return within 10-15 seconds.
- Page transitions should be handled by Next.js optimized routing (under 2 seconds). ##### 2.4.2 Safety Requirements Safe handling of user authentication tokens via Supabase Auth. ##### 2.4.3 Security Requirements
- Row Level Security (RLS) in the database ensures users only see their own notes.
- Sanitization of transcript input to prevents injection attacks.

Chapter No 3: Analysis (Use Case Model)

3.1 Identifying Actors and Use Cases

- **Primary Actor:** Learner (Student/User).
- **Secondary Actor:** AI Engine (Gemini).
- **System Actor:** Supabase Database.

3.2 Forming Use Case Diagram

- **Use Case 1:** User Sign-up/Login.
- **Use Case 2:** Input YouTube Link.
- **Use Case 3:** Generate Study Notes.
- **Use Case 4:** Take Quiz.
- **Use Case 5:** View Learning Analytics.

3.3 Describe the Events Flow for Use Case: “Generate Notes”

1. User enters URL in the “Process Video” tab.
2. System extracts video ID and checks DB for existing record.
3. If new, system fetches transcript.
4. User clicks “Generate Notes”.
5. AI processes transcript and returns Markdown.
6. System saves notes to Supabase and updates Dashboard.

Chapter No 4: Design

4.1 Architecture Diagram

Client-Server Architecture (Next.js 15): [Client Browser] <-> [Next.js Server API Routes] <-> [Supabase DB / Gemini AI]

4.2 ERD with Data Dictionary

- **Users:** id (UUID), email, name, avatar.
- **Videos:** id, user_id, title, youtube_url, transcript.
- **Notes:** id, video_id, user_id, content, is_bookmarked.
- **Learning Stats:** id, user_id, stats_date, total_minutes, session_count, current_streak.

4.3 Data Flow Diagram (DFD)

- **Level 0:** User sends URL -> NoteTube -> AI Result -> User.
 - **Level 1:** URL -> ID Extractor -> Transcript Fetcher -> AI Prompt -> DB Store -> UI Render.
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Chapter No 5: Implementation

5.1 Component Diagram

- **UI Base Components:** Buttons, Cards, Inputs.
- **Feature Components:** Header, Sidebar, DashboardLayout, NotesDisplay, QuizDisplay.

5.2 Deployment Diagram

- **Platform:** Vercel (Frontend + Serverless API).
- **Database:** Supabase Cloud.
- **AI Infrastructure:** Google Vertex AI / Gemini Platform.

5.3 Database Architecture

NoteTube uses a **3-Tier Architecture**: 1. **Presentation Layer:** Next.js Client Components. 2. **Logic Layer:** Next.js API Routes (Serverless). 3. **Data Layer:** PostgreSQL (via Supabase).

Chapter No 6: Testing (Quality Attributes)

6.1 Test Case Specification

Validation of core flows: Auth, Video Extraction, AI Generation, Persistence.

6.2 Black Box Test Cases

- **BVA:** Testing YouTube URLs with varying lengths; testing empty IDs.
- **Equivalence Partitioning:** Categorizing valid vs. invalid YouTube links.
- **Decision Table:** Authenticated vs. Guest user permissions for private pages.

6.3 White Box Testing

- **Statement Coverage:** Ensuring all API error handlings (`if (!user)`, `if (error)`) are executed during unit tests.
 - **Branch Coverage:** Testing both “Video Found in DB” and “Video is New” logic paths in `process-video/page.tsx`.
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Chapter No 7: Tools and Technologies

7.1 Programming Languages

- **TypeScript:** Ensuring type safety across the full stack.
- **JSX/TSX:** Component structure.
- **SQL:** Complex analytics queries in Supabase.

7.2 Operating Environment

- **Node.js:** Server environment.
- **Next.js 15:** Framework for SSR and client-side logic.
- **Tailwind CSS:** Responsive styling engine.